Atif Belal

+1-514-703-3367 | atif.belal.1@ens.etsmtl.ca | LinkedIn | GitHub | Montreal, Canada

Summary

Ph.D. candidate in Machine Learning at ÉTS Montréal with over six years of experience in artificial intelligence and computer vision, specializing in object detection, vision-language models, video anomaly prediction, multimodal LLMs, and domain adaptation. Published at top-tier conferences, including ECCV, ICCV, and WACV, with research experience at the University of Trento and industry experience at Ericsson.

Experience

Visiting PhD Researcher 07/2026 - 12/2026
University of Trento *Trento, Italy*

- **Video Anomaly Prediction with Localization:** Developing training-free methods that use multimodal LLMs to predict whether an anomaly will occur and which people will be involved.
- Built spatio-temporal annotations by tracking involved people backward from each anomaly with SAM3, refined through shot-cut detection, cycle-consistency checks, and manual review, and added per-video captions.
- Benchmarking multimodal LLMs as zero-shot baselines using frame-level AUC, temporal AP, and spatial AP.

Machine Learning Doctoral Research Assistant 05/2022 - Present
École de Technologie Supérieure *Montreal, Canada*

- **Vision-Language Object Detection:** Proposed the first test-time adaptation method for vision-language object detectors using weighted entropy minimization and image-conditioned prompt selection.
- Improved performance across multiple datasets by an average of 9.8 AP50 points while optimizing only 2.1% of the model parameters.
- **Domain-Adaptive Object Detection:** Proposed a method for unsupervised multi-source domain adaptation in object detection utilizing prototype and attention-based methods for class-level alignment.
- Improved performance on BDD100K by 8.1 AP50 points while reducing the number of trainable parameters by 50 million, leading to faster and more efficient computation.
- Demonstrated robustness in handling imbalanced datasets, achieving 6.8% improvement in imbalanced settings.

Mitacs Accelerate Intern 09/2022 - 03/2024
GAIA, Ericsson *Montreal, Canada*

- Developed an object detection model, enabling precise identification of antennas on towers from drone footage; reduced false positives by 35% under challenging weather scenarios.
- Trained the model in a semi-supervised setting using only 40% of the labeled data.
- Enhanced detection accuracy by 11% mAP compared to the previous state-of-the-art method.

Artificial Intelligence Research Engineer 04/2020 - 08/2021
LIVIA Laboratory, ETS *Montreal, Canada (Remote)*

- Collaborated with a doctoral candidate to develop a progressive multi-target domain adaptation method for image classification, improving generalization across multiple target domains by up to 15%.
- Enhanced real-time CNN applicability by reducing computational complexity by 30% while maintaining accuracy within 2% through combined knowledge distillation and unsupervised domain adaptation.
- Achieved state-of-the-art performance on benchmark datasets, improving accuracy by 4% on the OfficeHome dataset with the compressed model.

Selected Publications

- **DESA-TTA: Dynamic EMA and Source Anchoring for Test-Time Adaptation**
  ***Atif Belal**, et al., Submitted to the ICLR 2027*
- **VLOD-TTA: Test Time Adaptation of Vision-Language Object Detectors**
  ***Atif Belal**, et al., European Conference on Computer Vision (ECCV), 2026*
- **Multi-Source Domain Adaptation for Object Detection with Prototype-based Mean-teacher**
  ***Atif Belal**, et al., Proceedings of the IEEE/CVF Winter Conference on Applications of Computer Vision (WACV), 2024*
- **Visual Modality Prompt for Adapting Vision-Language Object Detectors**
  *Heitor R. Medeiros, **Atif Belal**, et al., International Conference on Computer Vision (ICCV), 2025*

Professional Skills

- **Programming Languages**: Python, C/C++, HTML, CSS, SQL
- **Machine Learning and Computer Vision**: scikit-learn, NumPy, Pandas, SciPy, Matplotlib, OpenCV, Detectron2
- **Deep Learning**: PyTorch, Keras, TensorFlow, Hugging Face, Transformers, LangChain, OpenAI API
- **Vision-Language**: SAM3, QWEN-VL, Grounding DINO, YOLO-World
- **Other Technical Skills**: Git, Linux, Azure, Google Cloud, Docker, Kubernetes
- **Soft Skills**: Communication, Presentation, Collaboration, Team Work, Creativity, Leadership, Management

Education

École de Technologie Supérieure 05/2022 - Present
Ph.D. in Machine Learning ***GPA: 4.3/4.3***

Aligarh Muslim University 08/2017 - 07/2021
Bachelor of Technology in Computer Engineering ***GPA: 4.05/4.3***